

1. What is Cloud Computing?

Cloud computing is the delivery of computing services including servers, storage, databases, networking, software, analytics, and intelligence over the Internet (“the cloud”).

Instead of buying, owning, and maintaining physical servers and data centres, organizations rent access to storage, databases, and computing power from a cloud provider on an as-needed basis.

Core Benefits:

- **Cost Efficiency:** Eliminates the capital expense of buying hardware and software. You operate on a pay-as-you-go model.
- **Speed & Agility:** Vast amounts of computing resources can be provisioned in minutes, giving businesses immense flexibility.
- **Global Scale:** Ability to scale resources elastically delivering the right amount of IT resources (e.g., more or less computing power, storage, bandwidth) right when they are needed and from the right geographic location.
- **Productivity:** Removes the need for many IT management chores, such as racking and stacking hardware, software patching, and other time-consuming IT maintenance.
- **Reliability:** Makes data backup, disaster recovery, and business continuity easier and less expensive because data can be mirrored at multiple redundant sites.

2. What are the Advantages of Microsoft Azure?

Microsoft Azure is one of the leading global public cloud platforms. It offers unique benefits, particularly for enterprises:

1. **Hybrid Cloud Capabilities:** Azure excels at hybrid deployment. It seamlessly connects on-premises data centres with the Azure public cloud using tools like Azure Arc and Azure Stack, allowing companies to transition smoothly.
2. **Deep Microsoft Ecosystem Integration:** If your organization already uses Windows Server, Active Directory, SQL Server, or Microsoft 365, Azure integrates natively. This translates to reduced learning curves and cost-saving licensing benefits (such as Azure Hybrid Benefit).
3. **Advanced Security and Compliance:** Microsoft invests heavily in security and boasts a larger compliance portfolio (including GDPR, HIPAA, ISO 27001) than any other cloud provider, making it highly trusted by governments and financial institutions.
4. **Global Reach:** Azure has more global data centre regions than many other cloud competitors, enabling businesses to deploy applications closer to their global user base for reduced latency.
5. **AI and Innovation:** Azure provides industry-leading AI tools, machine learning frameworks, and analytics services (like Azure OpenAI service) that allow organizations to rapidly build intelligent applications.

3. What is Hosting in Azure?

Hosting in Azure refers to the practice of deploying your digital assets websites, web apps, databases, background services, APIs, and enterprise systems onto Microsoft's cloud infrastructure.

Instead of hosting a website on a server sitting in your office closet, you deploy it to Azure's remote, high-performance data centres.

Common Ways to Host in Azure:

- **Azure App Service (PaaS):** Ideal for hosting web applications and APIs. You upload your code (built in .NET, Java, Node.js, Python, etc.), and Azure handles the underlying scaling, security patching, and server management.
- **Azure Virtual Machines (IaaS):** If your software requires custom configurations or a legacy OS setup, you host it inside a dedicated virtual machine where you have complete admin/root access.
- **Azure Kubernetes Service (AKS):** Designed for modern microservices architectures, allowing you to host containerized applications efficiently.
- **Azure Static Web Apps:** Optimized for modern frontend frameworks (React, Angular, Vue) and static sites with global content delivery network (CDN) acceleration.

4. What is IaaS, PaaS, and SaaS?

Cloud computing is divided into three primary service categories. They represent different levels of control and responsibility between you and the cloud provider.

IaaS (Infrastructure as a Service)

- **Definition:** You rent IT infrastructure servers, virtual machines (VMs), storage, networks, and operating systems from a cloud provider on a pay-as-you-go basis.
- **Analogy:** Renting an empty plot of land. You have full freedom to build whatever house you want, but you are responsible for the construction and maintenance.
- **Example:** Microsoft Azure Virtual Machines.

PaaS (Platform as a Service)

- **Definition:** The cloud provider supplies an environment for building, testing, delivering, and managing software applications. It is designed to make it easier for developers to create web or mobile apps without worrying about setting up or managing the underlying infrastructure of servers, storage, network, and databases.
- **Analogy:** Renting an unfurnished apartment. The structural maintenance is handled by the landlord, but you bring your own furniture (code and data).
- **Example:** Azure App Services, Azure SQL Database.

SaaS (Software as a Service)

- **Definition:** A method for delivering software applications over the internet, on demand, and typically on a subscription basis. The cloud provider hosts and manages the software

application and underlying infrastructure, and handles any maintenance, like software upgrades and security patching.

- **Analogy:** Staying in a hotel room. Everything is provided and maintained for you; you just use the room.
- **Example:** Microsoft 365, Salesforce.